

Aussie Primary Academy | Australian Curriculum v9.0 | Learn, Investigate and Explain

Name: _____

Date: _____

LEARNING INTENTION	compare findings, evaluate fairness, draw evidence-based conclusions and ask a follow-up question
KEY IDEA	A conclusion answers the question with evidence. Evaluation checks fairness, consistency and possible error, then recommends a specific improvement or next investigation.
WORKED MODEL	Use the inquiry cycle: question -> prediction -> safe method -> observations/data -> evidence-based conclusion -> evaluation -> communication.

SUCCESS CRITERIA - I CAN

[] use the inquiry skill [] record evidence clearly [] evaluate quality [] explain a next step

A. BUILD THE SCIENCE IDEA

1. Match claim, evidence, reasoning, limitation and conclusion.

2. Identify whether each problem affects fairness, accuracy or reliability.

3. Explain why two groups may obtain different results.

B. APPLY AND EXPLAIN

4. Write a conclusion that cites two numerical results.

5. Name one specific error and explain its likely effect.

6. Compare findings fairly without assuming one group is careless.

C. INDEPENDENT CHECK

7. Use a checklist to judge controls, repeats, scale and recording.

8. Write one precise improvement and one new follow-up question.

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D. APPLY SCIENCE IN CONTEXT

9. Scenario: three groups test which soil drains fastest and report similar but not identical times

10. Decide what the combined evidence supports and what it cannot show.

11. Write a conclusion with evidence, limitation and next step.

E. REASON, EVALUATE AND TRANSFER

12. Correct: Our one trial proved the prediction forever.

13. Explain how a fairer method could change confidence even if the result stayed similar.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

[] I can compare findings, evaluate fairness, draw evidence-based conclusions and ask a follow-up question.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: _____

The most difficult science idea was _____ because _____

One correction or improvement I made: _____

A question I still have: _____

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Accept equivalent responses when the student uses accurate science ideas, relevant evidence and clear reasoning.

COMPLETE ANSWER KEY

1. Claim answers; evidence is data; reasoning links; limitation bounds; conclusion synthesises.
2. Fairness=unequal variables; accuracy=measurement closeness; reliability=consistency/repeats.
3. Methods, controls, samples, instruments or natural variation may differ.
4. Must answer the question and cite values.
5. Name the cause and likely direction/effect, not just human error.
6. Compare method and evidence respectfully.
7. All four quality features should be checked.
8. Improvement is actionable; follow-up extends the investigation.
- 9-11. Accept a cautious evidence-based conclusion for the scenario.
12. One trial provides limited evidence; it does not prove forever.
13. Better controls/repeats reduce alternative explanations and increase confidence.

TEACHER CHECK AND NEXT STEP

[] Uses accurate vocabulary and a complete cause-and-effect explanation.

[] Supports conclusions with observations or data.

Next step: revisit one misconception, then transfer the idea to a new example.

COMMON MISCONCEPTION / WATCH OUT

Do not write only 'human error'; identify the exact problem and its effect.

A conclusion must stay proportional to the amount and quality of evidence.

EXTENSION AND RE-TEACH

Re-teach: model one difficult item, complete one together, then ask the student to explain an independent correction.

Extension: apply the science idea to a new real-world example and justify the explanation with observations or data.